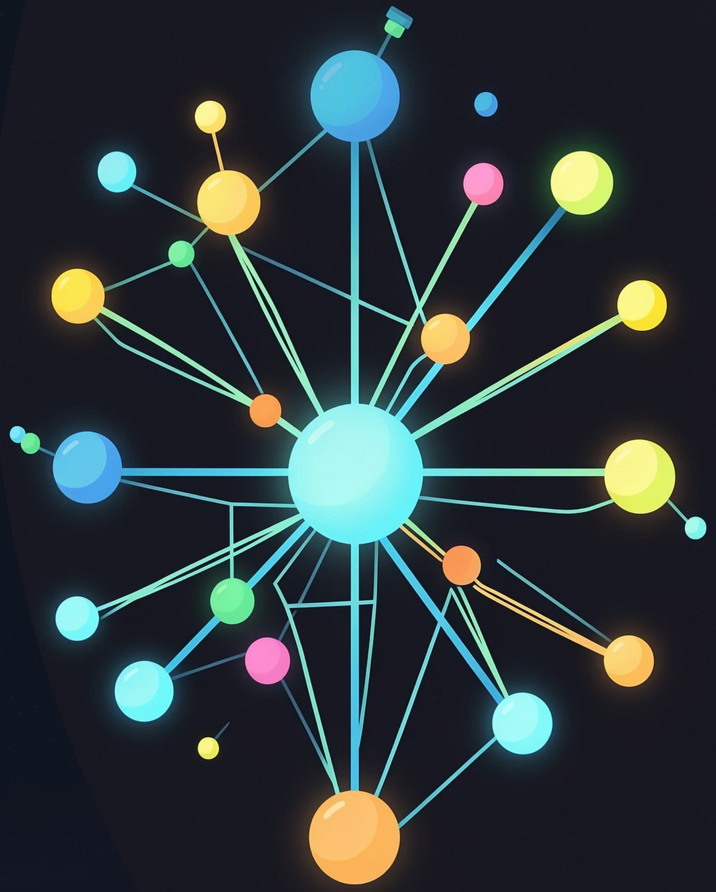
Customer Transaction Prediction

A machine learning project to predict whether a customer will make a transaction based on anonymized customer features — exploring model performance, feature importance, and business recommendations.

MACHINE LEARNING | CLASSIFICATION | PROJECT SUMMARY



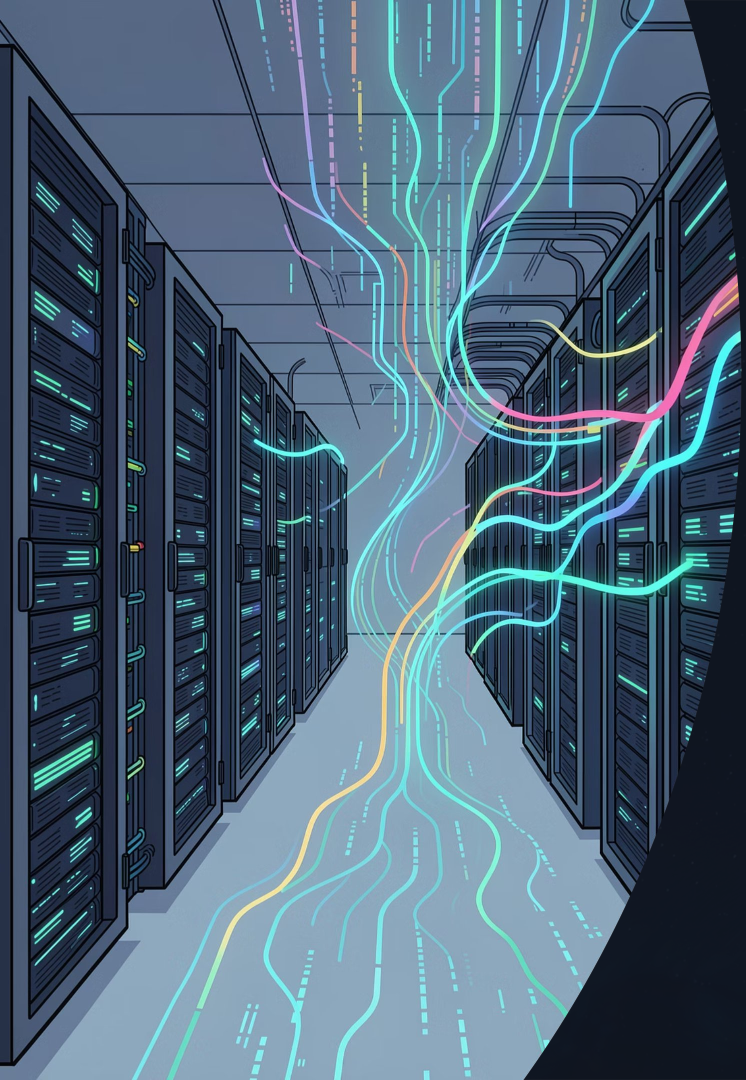
Project Objective

The Goal

Develop a machine learning model to predict whether a customer will make a transaction based on anonymized customer features. The project evaluates multiple classification algorithms, compares their performance, and recommends the best model for production deployment.

Core Questions

- Can we accurately predict transaction behavior from anonymized features?
- Which model delivers the best test accuracy and balanced classification performance?
- Which features are the most influential predictors?



Dataset Overview

200K

Records

Total customer transaction records in the dataset

202

Columns

Features including ID, target variable, and 200 anonymized variables

200

Anonymized Features

Numerical variables var_0 through var_199

The dataset contains an ID column, a binary target variable indicating whether a transaction occurred, and 200 anonymized numerical features (var_0 to var_199). The anonymized nature of the features means feature engineering relies on statistical patterns rather than domain-specific interpretation.

Data Preprocessing

Validate Data

Check missing values and duplicates

Split Variables

Separate features from the target

Train/Test Split

80% training and 20% testing

Prepare Data

Scale, encode, and finalize inputs

The preprocessing workflow ensures data quality before model training. Missing values and duplicate records were checked to confirm data integrity. Features were then separated from the target variable, and the dataset was split into an 80% training set and a 20% testing set. This standard 80/20 split provides sufficient data for training while reserving enough samples to evaluate generalization performance on unseen data.

Models Implemented

Four classification models were trained and evaluated, progressing from simple to more complex approaches. Logistic Regression served as the baseline, while tree-based models explored non-linear patterns in the data.

1

Logistic Regression

A linear classifier serving as the baseline model, well-suited for binary classification tasks.

2

Decision Tree Classifier

A non-linear model that splits data based on feature thresholds, capable of capturing complex patterns.

3

Random Forest Classifier

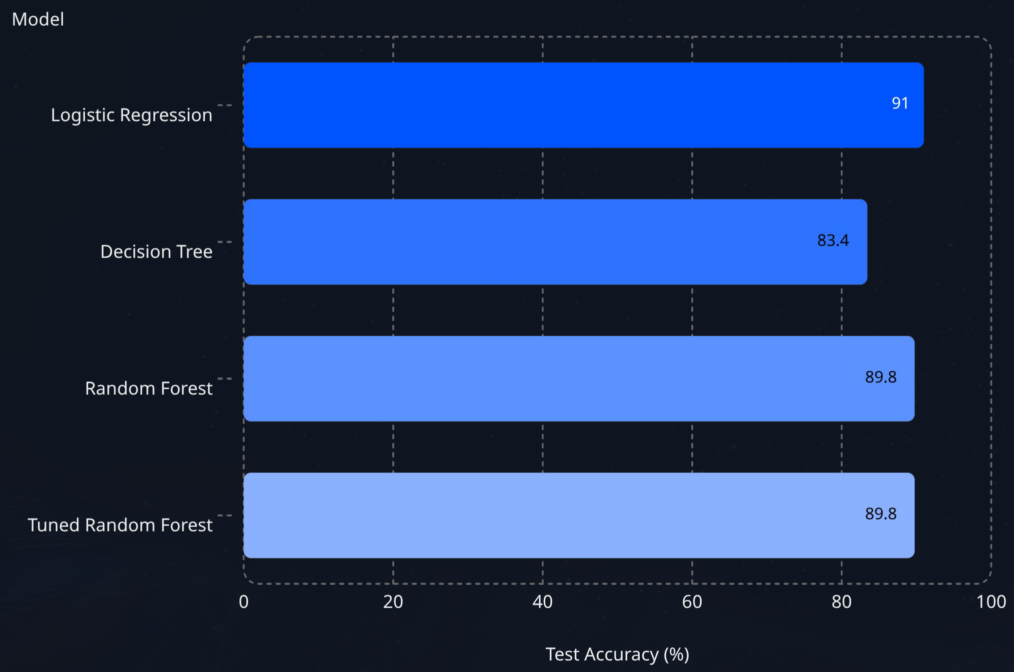
An ensemble of decision trees that improves stability and reduces overfitting through aggregation.

4

Tuned Random Forest

Random Forest optimized using GridSearchCV to find the best hyperparameter combination.

Model Performance



Key Results

Logistic Regression achieved the highest test accuracy at 91.00%, outperforming all tree-based models.

Decision Tree scored the lowest at 83.43%, showing signs of overfitting with 100% training accuracy but significantly lower test accuracy.

Random Forest and **Tuned Random Forest** both achieved 89.76% accuracy, with the tuned model reaching a ROC-AUC of 0.8203. GridSearchCV did not improve accuracy over the default Random Forest, suggesting the default hyperparameters were already near-optimal for this dataset.

Key Findings

Logistic Regression Leads

Logistic Regression achieved the highest overall test accuracy at 91.00%, outperforming all tree-based models despite its simplicity. It also demonstrated more balanced classification performance across both classes.

Decision Tree Overfitting

The Decision Tree showed clear signs of overfitting — achieving 100% training accuracy but only 83.43% on the test set. This gap indicates the model memorized training patterns rather than learning generalizable rules.

Class Imbalance Challenge

Random Forest improved stability over the single Decision Tree but struggled to identify positive-class transactions due to class imbalance in the dataset. Both Random Forest variants achieved identical 89.76% test accuracy.

Feature Importance

Top Predictors (Random Forest)

Random Forest feature importance analysis identified the following variables as the most influential predictors of customer transaction behavior:

- var_81
- var_12
- var_139
- var_53
- var_110

Why It Matters

Feature importance analysis helps identify which anonymized variables drive predictions the most. Understanding these key predictors can guide future feature engineering efforts and provide interpretability for business stakeholders, even when the original feature meanings are anonymized.

These five variables were the strongest signals in the Random Forest model for predicting transaction behavior.

Business Conclusion

The project successfully predicts customer transaction behavior using machine learning. After evaluating four models, **Logistic Regression is the recommended model** for production deployment because it delivered the best test accuracy and more balanced classification performance compared with tree-based models.

Best Accuracy

91.00% test accuracy — highest among all models evaluated

Balanced Performance

More consistent classification across both positive and negative classes

Production-Ready

Simple, interpretable, and efficient for real-world deployment

Future Improvements

Several avenues exist to further improve model performance and address current limitations:

Address Class Imbalance

Apply SMOTE (Synthetic Minority Over-sampling Technique) or class weighting to improve detection of positive-class transactions, which the current model struggles with.

Advanced Feature Engineering

Create derived features, interaction terms, and dimensionality reduction techniques to extract more signal from the 200 anonymized variables.

Hyperparameter Optimization

Expand GridSearchCV with broader parameter ranges and consider randomized or Bayesian search for more efficient tuning.

Boosting Algorithms

Evaluate gradient boosting frameworks such as **XGBoost**, **LightGBM**, or **CatBoost**, which often outperform Random Forest on tabular data.